ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

MATHEMATICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 7 Mathematics practice exam 1.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct methods.

QUESTION 1

1 1.1 C ✓ 1.2 C ✓ 1.3 A ✓ 1.4 B ✓ 1.5 C ✓ 1.6 B ✓ 1.7 D ✓ 1.8 C ✓ 1.9 C ✓ 1.10 B ✓ [10]

QUESTION 2

2.1 (a) 38 650 ✓ (b) 39 000 ✓ (2)

2.2.1 $9\,004 - 3\,678$ (column method with borrowing) ✓ $= 5\,326$ ✓ (2)

2.2.2 $347 \times 20 = 6\,940$ ✓ $347 \times 6 = 2\,082$ ✓ $6\,940 + 2\,082 = 9\,022$ ✓ (3)

2.2.3 Long division: $28 \div 12 = 2$ r 4 ✓; $45 \div 12 = 3$ r 9; $96 \div 12 = 8$ ✓ $\rightarrow 238$ ✓ (3)

[10]

QUESTION 3

3.1 5^3 ✓ (1)

3.2 (a) $27 + 16$ ✓ $= 43$ ✓ (b) 4×5 ✓ $= 20$ ✓ (4)

3.3 $18 = 2 \times 3 \times 3$ ✓ $24 = 2 \times 2 \times 2 \times 3$ ✓ (4)
$\text{HCF} = 2 \times 3 = 6$ ✓ $\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 = 72$ ✓

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QUESTION 4

4.1.1 $^8/_{12} + {}^3/_{12}$ ✓ $= {}^{11}/_{12}$ ✓ (2)

4.1.2 $^7/_2 - {}^7/_4$ ✓ $= {}^{14}/_4 - {}^7/_4$ ✓ $= {}^7/_4 = 1\,{}^3/_4$ ✓ (3)

4.1.3 $^{60}/_{80}$ (or cancel first: $^{1\times3}/_{1\times4}$) ✓ $= {}^3/_4$ ✓ (2)

4.2 (a) 16,15 ✓ (b) 0,24 ✓ (c) 48 tenths $\div 6 = 8$ tenths ✓ $= 0{,}8$ ✓ (4)

4.3 $^3/_8 = 0{,}375$ and $45\% = 0{,}45$ ✓ $\rightarrow {}^3/_8$; 45% ; 0,5 ; 0,52 ✓ (2)

4.4 25% of R480 $= {}^1/_4 \times 480$ ✓ $=$ R120 ✓ Sale price $=$ R480 $-$ R120 $=$ R360 ✓ (3)

4.5 $^{42}/_{60} \times 100$ ✓ $= 70\%$ ✓ (2)

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QUESTION 5

5.1 (a) -5 ✓ (b) 14 ✓ (c) -21 ✓ (3)

5.2 $13 - (-4)$ ✓ $= 17\ ℃$ ✓ (2)

5.3 $-7\ ;-2\ ;\ 0\ ;\ 3\ ;\ 5$ ✓ (1)

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QUESTION 6

6.1.1 13 ✓ (1)

6.1.2 Number of matchsticks $= 3n + 1$ ✓✓ (or "multiply the pattern number by 3 and add 1") (2)

6.1.3 $3 \times 20 + 1 = 61$ ✓ (1)

6.1.4 $3n + 1 = 100 \rightarrow 3n = 99$ ✓ $\rightarrow$ Pattern 33 ✓ (2)

6.2 (a) 9 ✓ (b) 13 ✓ (c) 8 ✓ (3)

6.3 $5(4) - 2(3) = 20 - 6$ ✓ $= 14$ ✓ (2)

6.4 (a) $x = 7$ ✓ (b) $x = 9$ ✓ (c) $3x = 15$ ✓ $\rightarrow x = 5$ ✓ (4)

[15]

QUESTION 7

7.1 $x = 180° - 58° = 122°$ ✓ (angles on a straight line add up to 180°) ✓ (2)

7.2 $x = 47°$ (vertically opposite angles) ✓ $y = 180° - 47° = 133°$ ✓ (2)

7.3 $\angle R = 180° - (64° + 48°)$ ✓ $= 68°$ ✓ (the angles of a triangle add up to 180°) ✓ (3)

7.4 $\angle L = \angle M$ (base angles of an isosceles triangle) $\rightarrow \angle L = (180° - 40°) \div 2$ ✓ $= 70°$ ✓ (2)

7.5 (a) Rhombus ✓ (b) Trapezium ✓ (2)

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QUESTION 8

8.1 (a) $P = 2(12 + 7)$ ✓ $= 38\ cm$ ✓ (b) $A = 12 \times 7$ ✓ $= 84\ cm^2$ ✓ (4)

8.2 $A = ½ \times 10 \times 6$ ✓ $= 30\ cm^2$ ✓ (2)

8.3 Side $= 36 \div 4 = 9\ cm$ ✓ $A = 9 \times 9 = 81\ cm^2$ ✓ (2)

8.4 (a) $V = 5 \times 4 \times 3$ ✓ $= 60\ cm^3$ ✓ (5)
(b) $SA = 2(5 \times 4) + 2(5 \times 3) + 2(4 \times 3)$ ✓ $= 40 + 30 + 24$ ✓ $= 94\ cm^2$ ✓

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QUESTION 9

9.1.1 $9\ ;\ 9\ ;\ 11\ ;\ 12\ ;\ 14\ ;\ 15\ ;\ 15\ ;\ 15\ ;\ 16\ ;\ 18\ ;\ 20$ ✓ (1)

9.1.2 15 ✓ (1)

9.1.3 15 (the 6th value) ✓ (1)

9.1.4 $154 \div 11$ ✓ $= 14$ ✓ (2)

9.1.5 $20 - 9 = 11$ ✓ (1)

9.2 $\mathrm{P(blue)} = {}^{5}/_{10}$ ✓ $= {}^{1}/_{2}$ ✓ (2)

[8]

TOTAL: 100

END OF MARKING GUIDELINE